

Srijana Shrestha

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SUMMARY

PhD candidate in Electrical Engineering specializing in **power and energy systems**. Skilled in **modeling, simulation, data analysis, and experimental validation** with applications in grid flexibility, demand response, and energy storage technologies. Interested in exploring control and data-driven techniques to solve challenges in grid modernization.

EDUCATION

University of Oklahoma, PhD in Electrical Engineering	2021 - 2026 (expected)	Norman, OK, USA
University of Oklahoma, MSc in Electrical Engineering	2021 - 2024	Norman, OK, USA
Tribhuvan University, Bachelor's in Electrical Engineering	2013 - 2017	Kathmandu, Nepal

SKILLS

Modeling & Simulation	MATLAB, Simulink, Simscape, ETAP, EMTP-RV, pandapower
Control & Automation	NI LabVIEW, NI hardware integration, Siemens PLC (TIA Portal), Ladder Logic, Functional Block Diagrams
Programming & Data	MATLAB, Python (pandas, scikit-learn), R, Statistical Regression & Machine Learning
Optimization	CPLEX, Linear Programming, MILP, Metaheuristics

PUBLICATIONS

- S. Shrestha, P. Moses and J. Cai, “**Application of a Squirrel Cage Induction Machine as a Dynamometer for Motor Heat-run Tests**,” *2024 IEEE ANDESCON*, Cusco, Peru, 2024. doi:10.1109/ANDESCON61840.2024.10755786
- S. Shrestha, V. Panchalogaranjan and P. Moses, “**The February 2021 U.S. Southwest Power Crisis**,” in *Electric Power Systems Research*, 2023. DOI:10.1016/j.epsr.2023.109124

EXPERIENCE

University of Oklahoma	Norman, OK, USA
Graduate Research Assistant	August 2021 - Present

- Implemented a cost-effective motor dynamometer setup using an induction machine by utilizing dynamic braking technique to achieve a precise and controlled loading with a control accuracy >95%.
- Built a fully automated laboratory testbed for insulation aging research by successfully integrating Siemens PLCs (for control) and NI LabVIEW (for control data acquisition).
- Developed an electromechanical model (accuracy >95%) of an induction machine in MATLAB/Simscape to quantify aging impact of speed modulation on the stator winding.
- Conducted a comparative analysis of Oklahoma and Texas electric power grids to identify key vulnerabilities under severe weather conditions.

Renewable Energy for Rural Livelihood	Kathmandu, Nepal
Electrical Engineer	June 2018 - July 2021

- Designed and supported implementation of 12 solar photovoltaic projects, including 10 utility mini-grids and 2 water pumping systems.
- Led end-to-end rehabilitation of a lightning-damaged solar project from identifying root cause and solution to procuring equipment and overseeing final installation.

PROJECTS

Battery Energy Storage Optimization

- Implemented a two-step approach using the Particle Swarm Optimization (PSO) algorithm in Python (pandapower library) for the optimal placement and sizing of a Battery Energy Storage System (BESS) in an electrical network.
- Demonstrated that optimum BESS placement on IEEE 14-bus and 30-bus test systems significantly reduced power losses by 7.5%, proving that BESS at the highest-load bus provides the most significant grid efficiency gains from local energy support.

LabVIEW-based Power Measurement System

- Designed a custom LabVIEW application to perform real-time acquisition, processing, and data logging of real, reactive, and apparent power for a variable frequency drive (VFD) fed induction machine.
- Ensured accuracy and reliability of the application by implementing methodologies in strict compliance with IEEE standards, adding a validated tool for future experimental analysis.

Temperature Control and Automation

- Designed a closed-loop automation system using a PLC and Ladder Logic for a temperature controlled environment. Used cost-effective sensors and implemented safety features, real-time monitoring, and data logging using Click PLC.
- Implemented a cost-effective hardware by modeling an expensive thermal sensor with a potentiometer.

Data Driven Temperature Estimation

- Investigated feasibility of linear regression models for real-time temperature estimation in a Permanent Magnet Synchronous Machine (PMSM) to prevent overheating and failure.
- Demonstrated that linear models were insufficient by achieving a max R^2 value of 0.29 for the multivariable model which highlighted the motor's thermal behavior dominated by complex non-linear relationships.

Electricity Demand Forecasting

- Developed a monthly forecasting model (seasonal ARIMA(0,1,2)(0,1,1)[12]) by achieving a high-accuracy MAPE of <5% to support decision making in long term operational and resource planning.
- Developed a different daily forecasting model using dynamic regression to capture the complex non-linear effects of temperature and weekday/weekend patterns on short-term demand.

KEY COURSES

Core & Electives

Power System Protection, Control Theory, Digital Signal Processing, Linear Systems Analysis, Power Quality and EMI, Measurement and Automation